

Class notes - MTH 346/646 - 9/18/25

This file has notes about how to use Magma to do computations with the stuff we've talked about so far in MTH 346/646.

In Magma, every line has to end with a ;
Every assignment is done with := . For example:

```
> a := 4;
> b := 2;
> a/b;
2
```

Now, Magma knows that $a/b = 2$, but because division is involved, Magma is thinking of a/b as a rational number.

```
> Factorization(a/b);
```

```
>> Factorization(a/b);
      ^
```

Runtime error in 'Factorization': Bad argument types
Argument types given: FldRatElt

Since a/b is a rational number (Magma says it has type FldRatElt) it won't compute the prime factorization. We need to tell Magma to think of a/b as an integer.

```
> Z := Integers();
> c := Z!(a/b);
```

The expression $Z!(a/b)$ tells Magma to think of (a/b) as an element of Z . This "coersion" of telling Magma to take an object of one type and think of it as another type is very common.

```
> c;
2
```

Now it will factor it.

```
> Factorization(c);
[ <2, 1> ]
```

Magma is good at doing complicated exact arithmetic.

```
> Factorization(2^60-1);
[ <3, 2>, <5, 2>, <7, 1>, <11, 1>, <13, 1>, <31, 1>, <41, 1>, <61, 1>, <151, 1>,
<331, 1>, <1321, 1> ]
```

Magma can tell you if a number is a square.

```
> IsSquare(1089);
true 33
> IsSquare(9801);
true 99
> chk, sqrroot := IsSquare(5234^3 + 17);
> chk;
true
> sqrroot;
378661
> F := GF(37);
```

Here F is $F_{37} = Z/37Z$ is the field of size 37.

```
> a := (F!2);
> a^18;
36
```

Of course 2^{18} isn't equal to 36, but $2^{18} = 36 \pmod{37}$.

```
> R<x> := PolynomialRing(Rationals());
> g := x^3 - 2*x + 3;
```

Magma can factor polynomials

```
> g := x^3 - 2*x + 1;
> Factorization(g);
[
  <x - 1, 1>,
  <x^2 + x - 1, 1>
]
```

Now let's work with some conic sections.

```
> P2<x,y,z> := ProjectiveSpace(Rationals(),2);
> f := x^2 + x*y + y^2 - 109*z^2;
> C := Conic(P2,f);
> BadPrimes(C);
[]
```

The set BadPrimes is the set of primes p we can use to show there are no solutions (by reducing mod p). In this case, the set is empty.

```
> HasRationalPoint(C);
true (5 : 7 : 1)
```

Magma finds a rational point on our conic.

```
> Evaluate(f,[5,7,1]);
0
> f := x^2 + x*y + y^2 - 107*z^2;
> C := Conic(P2,f);
> BadPrimes(C);
[ 3, 107 ]
```

This conic has bad primes,

```
> HasRationalPoint(C);
false
```

so it doesn't have any rational points.

```
> f := x^2 + x*y + y^2 - 109*z^2;
> P1<u,v> := ProjectiveSpace(Rationals(),1);
> C := Conic(P2,f);
```

Magma can parametrize the rational points on a conic.

```
> mp := Parametrization(C,Curve(P1));
> mp;
Mapping from: Curve over Rational Field defined by
0 to CrvCon: C
with equations :
5*u^2 + 24*u*v + 7*v^2
7*u^2 - 10*u*v - 12*v^2
u^2 + u*v + v^2
and inverse
2*x - 17*y - 218*z
-19*x - 2*y + 109*z
and alternative inverse equations :
-19*x - 2*y + 109*z
17*x + 19*y - 218*z
> C;
Conic over Rational Field defined by
x^2 + x*y + y^2 - 109*z^2
```

This mp is a map from P^1 to C . We can plug in some points

```
> mp([2,1]);
(75/7 : -4/7 : 1)
```

```
> mp([3,1]);
(124/13 : 21/13 : 1)
```

Let's check Magma's parametrization.

```
> X := 5*u^2 + 24*u*v + 7*v^2;
> Y := 7*u^2 - 10*u*v - 12*v^2;
> Z := u^2+u*v+v^2;
> X^2 + X*Y + Y^2 - 109*Z^2;
0
```

Now, let's work with some elliptic curves.

```
> E := EllipticCurve([0,0,0,-3,2]);
```

The list of five numbers here are numbers a_1 , a_2 , a_3 , a_4 , and a_6 so that $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$. Sometimes this longer form is needed to avoid problems at the primes 2 and 3. (Every curve $y^2 = x^3 + ax + b$ is singular mod 2, for example.)

This command gives a runtime error. It's because $y^2 = x^3 - 3x + 2$ is singular.

```
> E := EllipticCurve([0,0,0,14,1]);
```

This curve is $y^2 = x^3 + 14x + 1$.

```
> P1 := E![0,1,1];
> P2 := E![1,4,1];
> P3 := E![4,11,1];
> P1;
(0 : 1 : 1)
```

Let's have Magma double P_1 , that is, compute $P_1 + P_1$.

```
> 2*P1;
(49 : -344 : 1)
```

Let's have it do some other arithmetic.

```
> P1+P2;
(8 : -25 : 1)
> P1-2*P3;
(616 : -15289 : 1)
```

These were all integral points, because the x and y are integers. Here's a non-integral one.

```
> P1-3*P3;
(300001/93636 : -254255201/28652616 : 1)
> E2 := EllipticCurve([0,0,0,19,16]);
```

Let's search for points on $y^2 = x^3 + 19x + 16$.

```
> Points(E2 : Bound := 10);
{@ (0 : 1 : 0), (0 : 4 : 1), (0 : -4 : 1), (1 : 6 : 1), (1 : -6 : 1), (3 : 10 : 1), (3 : -10 : 1), (9/4 : 67/8 : 1), (9/4 : -67/8 : 1), (-5/9 : 62/27 : 1), (-5/9 : -62/27 : 1) @}

```

This searched for points where the numerators and denominators of the x coordinate are bounded by 10 (in absolute value).

```
> Points(E2 : Bound := 100);
{@ (0 : 1 : 0), (0 : 4 : 1), (0 : -4 : 1), (1 : 6 : 1), (1 : -6 : 1), (3 : 10 : 1), (3 : -10 : 1), (9/4 : 67/8 : 1), (9/4 : -67/8 : 1), (-5/9 : 62/27 : 1), (-5/9 : -62/27 : 1), (9/25 : 598/125 : 1), (9/25 : -598/125 : 1), (28 : 150 : 1), (28 : -150 : 1), (49/36 : 1439/216 : 1), (49/36 : -1439/216 : 1), (60 : 466 : 1), (60 : -466 : 1), (99 : 986 : 1), (99 : -986 : 1), (-71/100 : 1467/1000 : 1), (-71/100 : -1467/1000 : 1) @}
> Points(E2 : Bound := 1000);
```

```
{@ (0 : 1 : 0), (0 : 4 : 1), (0 : -4 : 1), (1 : 6 : 1), (1 : -6 : 1), (3 : 10 :
1), (3 : -10 : 1), (9/4 : 67/8 : 1), (9/4 : -67/8 : 1), (-5/9 : 62/27 : 1),
(-5/9 : -62/27 : 1), (9/25 : 598/125 : 1), (9/25 : -598/125 : 1), (28 : 150 :
1), (28 : -150 : 1), (49/36 : 1439/216 : 1), (49/36 : -1439/216 : 1), (60 : 466
: 1), (60 : -466 : 1), (99 : 986 : 1), (99 : -986 : 1), (-71/100 : 1467/1000 :
1), (-71/100 : -1467/1000 : 1), (124/81 : 5086/729 : 1), (124/81 : -5086/729 :
1), (129 : 1466 : 1), (129 : -1466 : 1), (169/9 : 2258/27 : 1), (169/9 :
-2258/27 : 1), (-159/196 : 631/2744 : 1), (-159/196 : -631/2744 : 1), (249/25 :
4318/125 : 1), (249/25 : -4318/125 : 1), (256/49 : 5508/343 : 1), (256/49 :
-5508/343 : 1), (355/81 : 9874/729 : 1), (355/81 : -9874/729 : 1), (361/64 :
8907/512 : 1), (361/64 : -8907/512 : 1), (595 : 14514 : 1), (595 : -14514 : 1),
(609/64 : 16657/512 : 1), (609/64 : -16657/512 : 1) @}

```

```
> E := EllipticCurve([1,2,3,4,5]);
> E;
Elliptic Curve defined by  $y^2 + x*y + 3*y = x^3 + 2*x^2 + 4*x + 5$  over Rational
Field

```

Let's have Magma put this in the form $y^2 = x^3 + ax + b$

```
> E2, mp, invmp := WeierstrassModel(E);
> E2;
Elliptic Curve defined by  $y^2 = x^3 + 4941*x + 185166$  over Rational Field

```

The command `WeierstrassModel` returns three things - the new curve,
a map from the old curve to the new curve, and a map from the new curve
to the old curve.

```
> Points(E2 : Bound := 100);
{@ (0 : 1 : 0), (63 : 864 : 1), (63 : -864 : 1) @}
> Points(E2 : Bound := 1000);
{@ (0 : 1 : 0), (63 : 864 : 1), (63 : -864 : 1), (-495/16 : 3321/64 : 1),
(-495/16 : -3321/64 : 1) @}

```

Let's name a point on the new curve,

```
> pt := E2![-495/16,3321/64,1];

```

and map it back to the old curve.

```
> invmp(pt);
(-103/64 : -233/512 : 1)
> pt2 := E2![63,864];
> invmp(pt2);
(1 : 2 : 1)

```

This is a point on the old curve.

```
> P := E![1,2];
> P;
(1 : 2 : 1)
> 2*P;
(-103/64 : -233/512 : 1)
> 3*P;
(12145/27889 : -22841754/4657463 : 1)

```

Let's map 3P to the new curve.

```
> mp(3*P);
(1190223/27889 : -3205753632/4657463 : 1)

```

Now, let's use Magma to find maps between a plane cubic and an elliptic curve

```
> R<x,y,z> := PolynomialRing(Rationals(),3);
> f := x^3 + y^3 - 6*z^3;
> C := Curve(ProjectiveSpace(Rationals(),2),f);
> C;
Curve over Rational Field defined by
 $\$.1^3 + \$.2^3 - 6*\$.3^3$ 

```

Something weird happened here. The dollar signs show up as place holders for variables in a polynomial ring. We haven't named the variables in the polynomial ring, so Magma doesn't know what to call them.

Let's have Magma search for points on C. This has a different command than for elliptic curves.

```
> PointSearch(C,10);
[ (-1 : 1 : 0) ]
```

There's one point, at infinity.

```
> PointSearch(C,100);
[ (17/21 : 37/21 : 1), (37/21 : 17/21 : 1), (-1 : 1 : 0) ]
```

There are a couple more that correspond to $(37/21)^3 + (17/21)^3 = 6$.

```
> PointSearch(C,1000);
[ (17/21 : 37/21 : 1), (-1 : 1 : 0), (37/21 : 17/21 : 1) ]
> PointSearch(C,10000);
[ (17/21 : 37/21 : 1), (37/21 : 17/21 : 1), (-1 : 1 : 0) ]
> PointSearch(C,100000);
[ (-1 : 1 : 0), (37/21 : 17/21 : 1), (17/21 : 37/21 : 1) ]
> PointSearch(C,1000000);
[ (-1 : 1 : 0), (17/21 : 37/21 : 1), (37/21 : 17/21 : 1) ]
```

We didn't find any new ones. How can we find more points on this curve?

```
> pt := C![-1,1,0];
> E, mp := EllipticCurve(C,pt);
```

This command creates a map to an elliptic curve.

```
> mp;
Mapping from: CrvPln: C to CrvEll: E
with equations :
1/2*$.3
1/4*$.1
$.1 + $.2
```

Is this a rational map or a morphism? The "base scheme" (or base locus) of a rational map is the set of points in the domain where the map is undefined.

```
> BaseScheme(mp);
Scheme over Rational Field defined by
$.3,
$.1,
$.1 + $.2,
$.1^3 + $.2^3 - 6*$.3^3
```

```
> RationalPoints(BaseScheme(mp));
{@ @}
```

This is the empty set. There are no rational points on the curve C where mp is not defined. (In fact, there are no complex points where C is undefined.) So mp is a morphism.

```
> E;
Elliptic Curve defined by  $y^2 - 1/4*y = x^3 - 1/48$  over Rational Field
```

This isn't the nicest elliptic curve

```
> E2, mp2, invmp2 := WeierstrassModel(E);
> E2;
Elliptic Curve defined by  $y^2 = x^3 - 243$  over Rational Field
```

That's better.

Let's build one map that sends points on C to points on E2

```
bigmap := mp*mp2;
```

Magma composes maps from left to right. So bigmap means "do mp to a point on C, and then apply mp2 to the resulting point on E."

Let's take the point $(17/21 : 37/21 : 1)$ and map it to E2.

```
> P := C![17/21,37/21,1];
> bigmap(P);
(7 : -10 : 1)

> chk, bigmapinv := IsInvertible(bigmap);
> chk;
> bigmapinv;
```

Is this a morphism, or just a rational map?

```
> RationalPoints(BaseScheme(bigmapinv));
{@ (0 : 1 : 0) @}
```

There is a point where bigmapinv isn't defined.

```
> Degree(BaseScheme(bigmapinv));
6
```

There are 6 points (counting multiplicity) where bigmapinv isn't defined. We will kindly ask Magma to extend this inverse to a morphism.

```
> bigmapinv := Extend(bigmapinv);
> bigmapinv;
Mapping from: CrvEll: E2 to CrvPln: C
with equations :
1/3*$.1^2*$.2 + 9*$.1^2*$.3
-1/3*$.1^2*$.2 + 9*$.1^2*$.3
$.2^2*$.3 + 243*$.3^3
and inverse
18*$.3
27*$.1 - 27*$.2
$.1 + $.2
and alternative equations :
1/54*$.1^3 - 1/54*$.2^2*$.3 + 1/3*$.2*$.3^2 + 9/2*$.3^3
-1/3*$.2*$.3^2 + 9*$.3^3
$.1*$.3^2

27/2*$.1^3 + 1/3*$.2^3 - 9/2*$.2^2*$.3 - 6561/2*$.3^3
-1/3*$.2^3 + 9*$.2^2*$.3
$.1*$.2^2
```

We now have 3 sets of defining equations for the inverse map.

```
> Degree(BaseScheme(bigmapinv));
The map bigmapinv is now defined everywhere on E2 and so it is a morphism.
```

Now, on E2, we have some points:

```
> Points(E2 : Bound := 1000);
{@ (0 : 1 : 0), (7 : 10 : 1), (7 : -10 : 1) @}
> bigmapinv([0,1,0]);
(-1 : 1 : 0)
> bigmapinv([7,10,1]);
(37/21 : 17/21 : 1)
> bigmapinv([7,-10,1]);
(17/21 : 37/21 : 1)
```

How could we find more points on $x^3+y^3 = 6z^3$?

```
> Q := E2![7,10,1];
> 2*Q;
(16009/400 : 2021723/8000 : 1)

> bigmapinv(2*Q);
```

```
(2237723/960540 : -1805723/960540 : 1)
```

This is another point on $x^3 + y^3 = 6z^3$!

```
> a := (2237723/960540);
> b := -1805723/960540;
> a^3+b^3;
6
```

Now, let's do an example where we map a curve in the form $y^2 = \text{quartic}$ to an elliptic curve.

```
> R<x> := PolynomialRing(Rationals());
> f := x^4 + x^3 + x^2 + x + 1;
> C := HyperellipticCurve(f);
> C;
Hyperelliptic Curve defined by  $y^2 = x^4 + x^3 + x^2 + x + 1$  over Rational Field
```

Magma makes this homogeneous by writing

$y^2 = x^4 + x^3z + x^2z^2 + xz^3 + z^4$

and saying that y has degree 2, while x and z have degree 1. This is called weighted projective space $P(1,2,1)$.

```
> Points(C : Bound := 10);
{@ (1 : -1 : 0), (1 : 1 : 0), (-1 : -1 : 1), (-1 : 1 : 1), (0 : -1 : 1), (0 : 1 : 1), (3 : -11 : 1), (3 : 11 : 1), (1 : -11 : 3), (1 : 11 : 3) @}
> Points(C : Bound := 100);
{@ (1 : -1 : 0), (1 : 1 : 0), (-1 : -1 : 1), (-1 : 1 : 1), (0 : -1 : 1), (0 : 1 : 1), (3 : -11 : 1), (3 : 11 : 1), (1 : -11 : 3), (1 : 11 : 3), (-11 : -101 : 8), (-11 : 101 : 8), (-8 : -101 : 11), (-8 : 101 : 11) @}
> pt := C![1,-1,0];
> E, mp := EllipticCurve(C,pt);
> E;
Elliptic Curve defined by  $y^2 + xy + 5/4y = x^3 - 3/2x^2 - 55/16x$  over Rational Field
> E2, mp2 := WeierstrassModel(E);
> E2;
Elliptic Curve defined by  $y^2 = x^3 - 4320x - 43200$  over Rational Field
> mp(C![3,11,1]);
(-1/4 : -3/2 : 1)
> mp2(mp(C![3,11,1]));
(-24 : -216 : 1)
```

So we used the point $(1 : -1 : 0)$ to map C to E . The image of the point $(3 : 11 : 1)$ on E is $(-24 : -216 : 1)$.

Magma can also find all of the integral points on an elliptic curve. (Sometimes this makes Magma really work.)

```
> E := EllipticCurve([0,0,0,0,1025]);
> E;
Elliptic Curve defined by  $y^2 = x^3 + 1025$  over Rational Field
> IntegralPoints(E);
[ (-10 : -5 : 1), (-5 : 30 : 1), (-4 : -31 : 1), (-1 : 32 : 1), (4 : 33 : 1), (10 : 45 : 1), (20 : 95 : 1), (40 : -255 : 1), (50 : 355 : 1), (64 : -513 : 1), (155 : 1930 : 1), (166 : -2139 : 1), (446 : -9419 : 1), (920 : 27905 : 1), (3631 : -218796 : 1), (3730 : 227805 : 1) ]
[ <(-10 : -5 : 1), 1>, <(-5 : 30 : 1), 1>, <(-4 : -31 : 1), 1>, <(-1 : 32 : 1), 1>, <(4 : 33 : 1), 1>, <(10 : 45 : 1), 1>, <(20 : 95 : 1), 1>, <(40 : -255 : 1), 1>, <(50 : 355 : 1), 1>, <(64 : -513 : 1), 1>, <(155 : 1930 : 1), 1>, <(166 : -2139 : 1), 1>, <(446 : -9419 : 1), 1>, <(920 : 27905 : 1), 1>, <(3631 : -218796 : 1), 1>, <(3730 : 227805 : 1), 1> ]
```

Magma only lists each x -coordinate once. So $(-10 : 5 : 1)$ is also an integral point on E .

Now, let's have Magma map a curve defined by two quadratic equations to an elliptic curve. For example, if $n = 98$, then $n = 2x^2$ (with $x = 7$), $n+1 = 99 = 11y^2$ (with $y = 3$), $n+2 = 100 = z^2$ (with $z = 10$). We can represent this as $11y^2 - 2x^2 = 1$, $z^2 - 11y^2 = 1$.

We make these equations homogeneous by setting
 $11y^2 - 2x^2 = w^2$ and $z^2 - 11y^2 = w^2$.

```
> R<x,y,z,w> := PolynomialRing(Rationals(),4);
> f1 := 11*y^2 - 2*x^2 - w^2;
> f2 := z^2 - 11*y^2 - w^2;
> C := Curve(ProjectiveSpace(Rationals(),3),[f1,f2]);
```

Let's search for points on C.

```
> PointSearch(C,10);
[ (7 : 3 : -10 : 1), (-7 : -3 : -10 : 1), (7 : 3 : 10 : 1), (-7 : -3 : 10 : 1),
(-7 : 3 : -10 : 1), (7 : -3 : -10 : 1), (-7 : 3 : 10 : 1), (7 : -3 : 10 : 1) ]
```

Hmmm. We already knew about those points. Are there other points we don't know about?

```
> PointSearch(C,100);
[ (7 : 3 : -10 : 1), (-7 : 3 : 10 : 1), (-7 : 3 : -10 : 1), (7 : -3 : 10 : 1),
(-7 : -3 : 10 : 1), (7 : -3 : -10 : 1), (7 : 3 : 10 : 1), (-7 : -3 : -10 : 1) ]
> PointSearch(C,1000);
[ (-7 : 3 : -10 : 1), (-7 : 3 : 10 : 1), (7 : -3 : 10 : 1), (7 : -3 : -10 : 1),
(7 : 3 : 10 : 1), (-7 : -3 : 10 : 1), (-7 : -3 : -10 : 1), (7 : 3 : -10 : 1) ]
> PointSearch(C,10000);
[ (-7 : -3 : 10 : 1), (7 : 3 : 10 : 1), (-7 : 3 : 10 : 1), (-7 : -3 : -10 : 1),
(7 : -3 : -10 : 1), (7 : -3 : 10 : 1), (-7 : 3 : -10 : 1), (7 : 3 : -10 : 1) ]
> PointSearch(C,100000);
[ (7 : 3 : -10 : 1), (7 : 3 : 10 : 1), (7 : -3 : -10 : 1), (-7 : -3 : -10 : 1),
(-7 : -3 : 10 : 1), (-7 : 3 : 10 : 1), (-7 : 3 : -10 : 1), (7 : -3 : 10 : 1) ]
```

We haven't found any yet.

```
> pt := C![7,3,10,1];
> E, mp := EllipticCurve(C,pt);
> E;
Elliptic Curve defined by  $y^2 = x^3 - 117600/24990001*x^2 - 15055998720000/624500149980001*x + 590315622400000000/15606259372500374970001$ 
over Rational Field
```

Ouch. That doesn't look good.

```
> E2, mp2 := WeierstrassModel(E);
> E2;
Elliptic Curve defined by  $y^2 = x^3 - 19518548797440000/624500149980001*x$  over
Rational Field
```

That's not much better either. Let's try the command "MinimalModel". This returns the equation with the "smallest discriminant", although it's not always in the form $y^2 = x^3 + Ax + B$.

```
> E2, mp2 := MinimalModel(E);
> E2;
Elliptic Curve defined by  $y^2 = x^3 - 484*x$  over Rational Field
```

Much better.

```
> bigmap := mp*mp2;
> Degree(BaseScheme(bigmap));
> bigmap := Extend(bigmap);
```

This last command takes a little while.

```
> bigmap(C![7,3,10,1]);
(0 : 1 : 0)

> P := bigmap(C![7,3,-10,1]);
> P;
(-539/25: 2541/125 : 1)
```

This gives us a nice point on the elliptic curve.

Let's map it back!

```
> chk, bigmapinv := IsInvertible(bimap);  
> chk;  
true  
> Degree(BaseScheme(bigmapinv));  
8
```

Let's extend it

```
> bigmapinv := Extend(bigmapinv);
```

Now let's find another point!

```
> 2*P;  
(24019801/44100 : -117624975299/9261000 : 1)
```

Let's map this back to the curve.

```
> bigmapinv(2*P);  
(-161208607/72029999 : -72088803/72029999 : -249705970/72029999 : 1)
```

So we found another point on C!

Q: Is there anything you'd like to know how to do in Magma but don't?